6. D-FMEA

6.1 Definition

Design failure mode and effect analysis (DFMEA) is a structured method that helps engineers understand the impact of potential risks associated with a design.[18] It helps determine how possible design failures may affect performance, safety and reliability.

The FMEA process considers three main factors:

1. Severity (S) – The potential impact of a failure on the system or user.
2. Occurrence (O) – The likelihood of the failure occurring.
3. Detection (D) – Assess how easy or difficult it is to detect the failure before it affects the system.

The Risk Priority Number (RPN) is calculated using the following formula:

*RPN = Severity * Occurrence * Detection*

A higher RPN indicates a higher level of risk, requiring immediate attention and appropriate corrective action.

RPN alone is not always enough, because it treats Severity, Occurrence, and Detection equally. In D-FMEA, some failures should receive attention mainly because their effect is severe, even if the RPN is not the highest. Action Priority (AP) helps show which failure modes need action first by considering Severity first, then Occurrence, then Detection.[19]

[18] Ansys. What is DFMEA? Ansys Blog. Published February 7, 2024. Accessed May 12, 2026. https://www.ansys.com/blog/what-is-dfmea
[19] Quality-One International. AIAG & VDA FMEA Handbook and Guide. Quality-One International. Accessed May 12, 2026.

6.2 D-FMEA Rating Scheme

Severity Scale		
Effect	**Criteria: Severity of Effect**	**Ranking**
Hazardous - Without Warning	May expose client to loss, harm or major disruption - failure will occur **without** warning	10
Hazardous - With Warning	May expose client to loss, harm or major disruption - failure will occur **with** warning	9
Very High	Major disruption of service involving client interaction, resulting in either associate re-work or inconvenience to client	8
High	Minor disruption of service involving client interaction and resulting in either associate re-work or inconvenience to clients	7
Moderate	Major disruption of service not involving client interaction and resulting in either associate re-work or inconvenience to clients	6
Low	Minor disruption of service not involving client interaction and resulting in either associate re-work or inconvenience to clients	5
Very Low	Minor disruption of service involving client interaction that does not result in either associate re-work or inconvenience to clients	4
Minor	Minor disruption of service not involving client interaction and does not result in either associate re-work or inconvenience to clients	3
Very Minor	No disruption of service noticed by the client in any capacity and does not result in either associate re-work or inconvenience to clients	2
None	No Effect	1

Table 19 : Severity Scale for Failure Modes and Effects.

Occurrence Scale			
Adapt as appropriate			
Probability of Failure	**Time Period**	**Per Item Failure Rates**	**Ranking**
Very High: Failure is almost inevitable	More than once per day	>= 1 in 2	10
	Once every 3-4 days	1 in 3	9
High: Generally associated with processes similar to previous processes that have often failed	Once every week	1 in 8	8
	Once every month	1in 20	7
Moderate: Generally associated with processes similar to previous processes which have experienced occasional failures, but not in major proportions	Once every 3 months	1 in 80	6
	Once every 6 months	1 in 400	5
	Once a year	1 in 800	4
Low: Isolated failures associated with similar processes	Once every 1 - 3 years	1 in 1,500	3
Very Low: Only isolated failures associated with almost identical processes	Once every 3 - 6 years	1 in 3,000	2
Remote: Failure is unlikely. No failures associated with almost identical processes	Once Every 7+ Years	1 in 6000	1

Table 20 : Occurrence Scale.

Detection Scale

Detection	Criteria	Ranking
Almost Impossible	No known controls available to detect failure mode	10
Very Remote	Very remote likelihood current controls will detect failure mode	9
Remote	Remote likelihood current controls will detect failure mode	8
Very Low	Very low likelihood current controls will detect failure mode	7
Low	Low likelihood current controls will detect failure mode	6
Moderate	Moderate likelihood current controls will detect failure mode	5
Moderately High	Moderately high likelihood current controls will detect failure mode	4
High	High likelihood current controls will detect failure mode	3
Very High	Very high likelihood current controls will detect failure mode	2
Almost Certain	Current controls almost certain to detect the failure mode. Reliable detection controls are known with similar processes.	1

Table 21 : Detection Scale.

Action Priority Scale

Priority	Criteria	Ranking
Low	Corrective action is not immediately required. The failure mode has a low effect, or the current controls are sufficient to manage the risk.	3
Medium	Corrective action should be considered. The failure mode has a moderate effect, or the currebr controls may not be fully sufficient.	2
High	Corrective action is required. The failure mode has a significant effect and the current controls are not sufficient to prevent or detect it.	1

Table 22: Action Priority Scale.

D-FMEA

Process/Product Name: Sunny Pro
Responsible: Group 3
Prepared By: Group 3
FMEA Date (Orig.): (Rev.):

Component	Design Function	Failure Mode	Potential Failure Effects	Potential Causes	S (1 - 10)	O (1 - 10)	D (1 - 10)	RPN	Recommended Action	Action Priority (1-3)
What is the process step, change or feature under investigation?	In what ways could the step, change or feature go wrong?	What is the failure mode?	What is the impact on the customer if this failure is not prevented or corrected?	What causes the step, change or feature to go wrong? (how could it occur?)					What are the recommended actions for reducing the occurrence of the cause or improving detection?	
Power Supply System										
ON/OFF switch	Turns the machine on/off when user presses switch	no function	Machine cannot be switched on/off; user cannot operate the system	Defective switch; broken contact; loose wire; damaged connector	5	3	2	30	Use reliable switch and verify wiring connection	3
Power Supply	Provides required electrical power to the machine	No function	Machine does not turn on; user cannot operate the system	Defective power supply; blown fuse; loose connector	8	2	5	80	Have to replace power supply unit	2
	Provides stable voltage to controller and heating system	Under function	Machine works weakly or resets; heating may not reach required temperature	Low output voltage; overloaded supply; voltage drop; weak connector	6	2	5	60	Select power supply with sufficient margin; Check voltage under load and add voltage monitoring	2
	Supplies correct voltage and current level	Over function	Overvoltage may damage controller, display, sensors, or heater control	Regulator failure; short circuit in feedback path	9	2	5	90	Add overvoltage protection and monitoring	1
Main MCU STM	Controls machine logic and coordinates all subsystems	no function	Main MCU doesn't start; motors, sensor, the display, buzzer doesn't work; machine can't operate	Defective MCU	8	3	1	24	Replace MCU	2
		Under function	Some outputs work but others fail; display may update incorrectly, buzzer may not warn, rotating plate may not move correctly	Firmware error; wrong I/O configuration	8	4	9	288	Replace MCU	2
		Intermittent function	MCU resets or freezes during operation; stepper driver receives unstable commands	unstable power supply; overheating	7	4	8	224	Get a voltage regulator and voltage monitoring	2

Internal

		Unintended function/Malfunction	MCU sends wrong message; motor, buzzer, display and others parts may activate at the wrong time	Software logic error; communication problem	5	3	9	135	Individual subsytems tested with code	3
Storage Unit and Breaking Mechanism System										
Egg Storage Unit	Stores eggs safely	Under function	Egg cracks and causes mess; user must stop	Rough loading by staff	2	4	2	16	Staff has to remove eggs and clean	3
	Keep eggs availabe for cooking	No function	Unit runs empty; staff must reload eggs; causes delay in cooking cycle	Staff forgets to load eggs; storage capacity too small	1	3	1	3	There will be a warning pop-out on the screen - accessed by the staff	3
Gate Actuators	Opens/closes gate to release egg at correct time	Over function	More than one egg is released; eggs collide, crack	Actuator timing error; control signal error; gate doesn't close fast enough	5	3	6	90	Add me mechanical stopper for single egg release; improve gate timing	2
		No function	Gate doesn't open; eggs will cram at the opening; machine cycle can't start	Actuator failure; no power signal	7	3	3	63	Give error message if gate doesn't happen	2
Flow sensors	Detect wheter egg has passed through system	No function	Machine doesn't detect egg; cycle will stop unexpectly because no egg detection	Sensor gets blocked	6	2	7	84	Have it in the service manual to clean the sensor periodically	2
Flip Actuator	Flips eggshell into eggshell storage	Intermittent function	Actuator does not full required range	Power goes below threshhold	7	3	2	42	Voltage Monitoring System on Touch Screen	2
Spear Solenoid	Drives the wedge apart	Under function	Egg shell is not opened properly; egg shell may enter egg white and yolk	Desynchronization with wedge and platform solenoid so the wedge isnt spread apart to open egg shell properly	6	3	4	72	Synchronization through checks within the logic code and flagging desynchronization	2
Return spring	Returns spear mechanism to starting position	No function	Mechanism stays open or stuck, next egg may jam or release incorrectly	Spring fatigue; corrosion	8	2	7	112	Long-term service checks	1
Platform + Wedge Solenoid	Moves front/back to release egg	No function	Platform doesn't move into correct position; egg may drop incorrectly or machine cycle stops	Solenoid plunger jammed	7	3	6	126	Add sealed food-safe solenoid; position senor	2
Thermal Insulation	Reduces heat; transfers to outher shell	Under function	Outer shell becomes too hot; electronics may overheat	Insulation too thin; poor ventilation	8	3	6	144	Use higher-rated insulation	1

. Internal

	Prevents steam from affecting internal parts	Unintended Function/Malfunction	Moisture build up inside insulation area; insulation performance decreases; condensation may form near electronic parts	Steam from cooking trapped inside housing; insulation absorbs moisture	8	3	7	168	Use moisture-resistant insulation, add ventilation	1
Axial Fan	cools internal electronics	No function	Fan stops, electronics overheat; machine shuts down; electrical components can fail; steam or smoke can come out of machine	Fan motor failure; dust/grease buildup; poor power supply	9	3	2	54	Add temperature sensor, protective filter, make fan/filter area easy to clean	1
	removes heat from enclosure	Under function	Hot air is not removed efficiently; heat builds up inside machine; outer shell becomes warmer; cooking performance reduces	Fan too small; poor fan placement; dust/grease buildup	6	5	5	150	Improve fan placement; make fan/filter area easy to clean	2
Ventilation Grille	allows airflow while blocking debris	Under function	airflow is restricted; machine overheats; cooking performance decreases	Grease buildup; blocked by user placement	7	4	6	168	Put machine in an open area; add "do not block vents" label	2
Vent Duct Channel	Guides hot air put of machine	Under function	Hot air isn't removed properly; heat builds up in machine	Duct too narrow; grease blockage	7	4	6	168	Smooth duct path; increase duct size	2
Step down converter	Converts 24V DC to low-voltage DC power	Over function	Current is too high; electronics fail; solenoid overheat; machine stops working	Wrong converter rating; poor converter quality	7	2	4	56	Use certified converter and high current protection	2
	provides stable and safe electrical power to the system	Unintended function/ malfunction	PSU provides unstable power, electrical noise, short circuit: system becomes unsafe	Overheating	7	2	4	56	Check fuse for easy detection	2
Solenoid Driver	Controls current to solenoid at the correct time	Unintended function	It activates at wrong time or stays on; egg mechanism moves incorrectly	software error; short circuit	7	3	7	147	Add position feedback by using an encoder	2
Oil System										
Oil Tank	Stores oil	Unintended function	Oil leaks into machine; causes mess and cleaning delay, oil might leak onto hot surfaces	Overfilling	4	3	2	24	fill-level marking	3
Peristaltic Pump	Pumps oil from tank to noozle	No function	No oil flow; conveyer stays dry; egg may stick or burn	Defective pump; pump might be jammed	7	3	3	63	No function	2

Internal

		Under function	Too little oil is pumped and sprayed; egg may stick or burn, cooking quality reduces	Pump wear; low voltage; tube compression loss	6	4	4	96	Test the pump flow rate during assembly, controll run time through controller	2
		Over function	Too much oil is pumped and sprayed; too much oil on conveyor; egg will be too greasy and there's a risk of smoke	Pump runs too long; wrong software timing	8	4	2	64	Calibrate spray time	1
Inline Check Valve	Prevents backflow and dripping when pump stops	Unintended function / malfunction	Oil drips; conveyer becomes too oily; causes mess or smoke	Valve stuck open; oil residue buildup	4	4	4	64	Use a food-grade check valve	3
Silicone Tubing	Transfers oil between tank, pump, and noozle using flexible food-grade tubing	Under function	Reduced or stopped oil flow completly; uneven spraying, oil could leak into machine	Kinked tube; loose connection; heat damage	4	4	6	96	Use food-grade high-temperature tubing, secure it with clamps, avoid sharp bends	3
Flat Fan Spray Nozzle	Distributes oil evenly	No function	No oil reaches conveyor; egg may stick or burn	Noozle blocked; dried oil residue	5	5	5	125	Make the nozzle removable for cleaning	2
		Under function	Distributes oil unevenly or less then intended; some areas get too much oil while others too little, reduces cooking quality	Nozzle clogging	4	5	5	100	Make the nozzle removable for cleaning	3
MOSFET Driver Module	Controls pump operation electronically from controlsystem	No function	Pump doesn't run; no oil is sprayed onto conveyor	Defective MOSFET, no control signal	8	2	9	144	Get a verefied MOSFET	2
Screen System										
Touch Controller	Detects and processes user touch input	No function	User cannot operate the product; device will appear frozen	MCU failure	8	3	2	48	Use Power-on self-test and touch input check	2
	Sends selected command to the microcontroller	Intermittent function	Touch is sometimes detected; selection may fail or needs repeated touching	Unstable voltage; poor calibration, loose connector	4	4	2	32	Have firm connection	3
Display Interface	Transfers display data and control signal to TFT LCD Display	No function	No display data is transferred; screen remains blank or frozen	Broken interface line, wrong pin connection, missing clock/data signal, connector fault	6	4	3	72	No action	2
TFT LCD Display	Displays graphical user interface and system information to user	No function	Blank screen; user can't see menu; machine won't work	Defective display; damaged cable; broken by force; dead pixels	7	4	1	28	Periodic pixel checks	2

. Internal

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Backlight Driver	Controls current supplied to the display backlight LEDs	No function	Display is dark or unreadable; user cannot clearly see menu	Driver IC defect, broken LED line	4	3	3	36	Check backlight power supply	3
	Limits backlight	Over function	Backlight too bright; higher	Output current too high,	2	3	2	12	Get a verified	3
Display MCU	Processes display data and controls the graphical user interface	No function	Display does not show menu, cooking time, temperature, or warning messages; user cannot operate or monitor the machine correctly	Defective MCU, code crashes	7	3	3	63	Test the MCU periodically in the surface	1
Transportation System										
Driver for Stepper Motor	Controls function of stepper motor (Rotation and direction)	No function	Motor may not move/ get stuck, vibration without rotation	Insufficient voltage, inconsistent wiring	5	4	2	40	Verifying wiring polarity and coil connections	2
		Overfunction	Extra unintended steps, excessive heating, overtorque	Inadequate cooling, high voltage inducing spikes/resonance	6	3	1	18	Monitor LEDS, auto idle reduction if overheating occurs	2
Stepper Motor	Stepper motor advances conveyor belt to position egg plates precisely under heaters for timed cooking cycles.	No function	Complete belt stoppage; Upstream egg plates backup into loading zone; Overflow spills onto kitchen floor	Bearing seizure from grease dry-out; Coil windings shorted/burnt from overcurrent; Shaft key shear from belt jam torque	9	4	3	108	Add current signature monitoring for bearing wear detection, avoid resonance speeds	1
		Under function	Plates stop short of heater zone; Eggs partially cooked/misaligned Home sensor false trips repeatedly	Resonance, Acceleration ramp could be too aggressive	7	6	5	210	Install Auto-tensioners, encoders for slip detection	2
		Overfunction	Motor stalls with burning odor; Holding torque lost suddenly; Conveyor freewheels backward;	Stall current from continuous duty; No airflow; Conduction from nearby heat block	9	5	2	90	Installing a heatsink/fan, temperature monitor	1
Roller Chain	Transmit mechanical power and synchronized motion from the drive motor to the conveyor bed	No function	Catastrophic sprocket disengagement; Egg plates crash into heater block; Chain debris contaminates eggs	Fatigue, overload shock, Sprocket tooth shear transfers load spike;	10	3	3	90	Using an oversized chain, shock absorbers on the belt	1
		Under function	Slip/jam, product misalignment, Rotation part stops	Shaft bending; wear; lack of lubrication	7	3	6	126	Use strong corrosion-resistant shaft; add bearing support; use a stainless chain (15k hour life)	2

. Internal

		Overfunction	Binding/ Heat buildup; Sprocket galling locks system; Friction heat chars nearby eggs; Chain welds to sprocket	Excess lubricant, misalignment	7	4	4	112	Precision sockets, specified lubricant	2
		Unintended function	Vibration/noise, socket damage	Loose fit, imbalance	5	5	6	150	Chain guides, retainers	3
		Intermittent function	Erratic speed, intermittent slips	Wear debris, contamination	6	5	5	150	Improved seals/filters	2
Plates on Conveyer	Trasports the eggs while they cook on the belt	No function	Raw eggs/ contamination	Heater burnout, poor contact	9	4	3	108	Redundant heaters, fault alarm system	1
		Under function	Undercooked/sticky eggs	Thermal warp, uneven heating	8	5	4	160	Aluminum core for conduction, insulation below	2
		Overfunction	Burnt/charred eggs, fire risk	Overcooked, stuck eggs	10	4	3	120	PID temp control, overheat fuse	1
Home Centering Sensor	Detects egg presence and centering in plates to halt conveyor if misaligned.	No function	Empty conveyor plates proceed, eggs may not fall direclty onto the plate	Dirt blockage, emitter could fail	8	5	4	160	Cleaning with air blasts, redundant sensors	2
		Under function	Unnecessary stops, downtime	Ambient light interference, misalignment	6	6	5	180	IR filter lens, adjustable threshold	3
		Over function	Off-center eggs proceed, breakage	Reflective residue, range overshoot	8	4	4	128	Dual-beam verification, anti-reflect coating	2
End position sensor	IR end position sensor detects last plate passage to trigger conveyor reset for first plate home positioning.	No function	Conveyor never resets, jam	Dust block, LED fail	9	5	4	180	Auto clean sensor, redundant IR	1
		Under function	Premature reset, position error	Ambient light, misalignment	5	6	5	150	IR filter, auto-calibrate	3
		Over function	Skips reset, endless cycle	Reflective dust, range error	8	4	4	128	Polarized lens, dual check	2
Egg-Shell Storage System										
Main waste drawer tray	Removable stainless steel drawer used for shell waste collection	No function	Egg shells are not collected; shells may fall into machine	Drawer not inserted correctly	7	3	2	42	Improve drawer positioning and add clear insertion guide	2
Linear Ball Bearings / Bushings	Supports smooth sliding motion of guide rods	No function	Pusher mechanism cannot move; egg shells are not compressed; drawer fills too early	Bearing seized; dirt/shell fragments inside bearing	4	3	3	36	Check sliding movement during assembly	3

. Internal

Return Springs	Returns pusher mechanism to default position after operation	No function	Pusher stays in compressed position; new shells may not enter correctly; drawer may jam	Broken spring	6	3	3	54	Replace spring	2
		Under function	Pusher returns slowly or only partly; shell compression mechanism may block next cycle	Spring fatigue	4	4	4	64	Use spring with high lifetime margin and test return force	3
Bottom Heating System										
Temperature Controller	Regulates heating temperature using feedback control	No function	Raw/safe eggs, contamination	Power fail, thermocouple disconnect	9	4	3	108	Watchdog timer, backup controller	1
		Under Function	Undercooked eggs, bacteria risk	Sensor drift, poor tuning	8	5	4	160	Calibrated RTD/thermocouple, dual sensors	2
		Over Function	Burnt eggs, fire hazard	Heater stuck-on, setpoint error	10	4	3	120	Independent safety cutoff, thermal fuse	1
Thermal fuse	Provides over-temperature protection for the heating system	No function	Uncontrolled overheat, fire	Defective fuse, wrong rating	10	3	2	60	High-reliability rated fuse, testing of fuse	1
		Under function	Frequent shutdowns, downtime	Poor thermal coupling to block	6	5	3	90	Optimized block embed position	2
		Overfunction	Excessive block temp before cutoff; causes extreme heating issues	High thermal mass lag between actual overheating and detection by fuse	9	4	3	108	Dual fuses, closer to cartridges	1
Solid State Relay	Switches cartridge heater power	No function (Off)	No heat, raw eggs	Input signal loss,	7	4	4	112	Dual SSR parallel, signal watchdog	3
		Overfunction	Overheat/burn, fire	Fail-short from surge	8	3	2	48	Included a Thermal fuse for safety	2
Cartridge Heaters	Tube-shaped, heavy-duty, industrial Joule heating element	No function	Cold block, raw eggs	Coil burnout, vibration wire break	8	4	4	128	Redundant cartridges, vibration damp	2
		Under function	Slow/uneven cooking	MgO moisture ingress, partial short	6	5	4	120	Tight bore fit (H7/g6), moisture seal	3
		Over function	Local burning, fire risk, hot spots	Poor fit/voids, overload	10	3	3	90	Multi-zone wiring, thermocouples	1
Aluminium Block	Conductive block with drilled heater/sensor holes	No function	Hot spot failure in the block, uneven cook	Drill stress riser, thermal shock	7	3	3	63	Ultrasonic test, pilot drills	2
		Under function	Poor thermal contact, slow heat	Oversize holes because thermal expansion	7	5	4	140	Precision ream H7, thermal compound	2
		Over function	Block edge overheating	Oversize holes because thermal expansion	6	4	5	120	Optimize hole spacing/depth	2
Temperature Sensor	Used to check the temperature of the aluminium block	No function	No temp feedback, overheat risk	Wire break, connector fails	9	4	3	108	Dual sensors voting, continuity monitor	1
		Under function	Undercooking from false cold	Sensor drift, poor thermal contact	7	5	4	140	Spring-loaded sensor, annual calibration	2

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		Over function	Overcooling, wasted energy	Insulation short, moisture ingress	4	4	4	64	Moisture-proof connector, dielectric test	3
TOP IR Heating System										
IR Heating element	400–450 mm shielded IR heater, 300–600 W class, heats eggs from above	No function	Undercooked tops, raw yolks	Filament burn, power loss	6	4	4	96	Redundant lamps, auto-switch	3
		Under function	Inconsistent cooking	Element aging, voltage drop	4	5	4	80	Voltage regulator, spare element	3
		Over function	Scorched tops, smoke/fire	Overdrive power, reflector focus	9	5	3	135	Current limiter, IR sensor	1
Rotating Plate System										
Stepper Driver Module	Controls stepping sequence of motor	No function	Driver doesn't work; Motor won't work; Rubber wheel doesn't move; Plate won't rotate; eggs will overlap and damage the yolk	Defective driver, no power supply, driver overheating	4	4	2	48	Check power/enable signal	3
		Under function	Motor receives weak/incorrect signal; plate moves too slowly; eggs may overlap and damage the yolk	Motor current too low, wrong step setting, mechanical friction, overload from plate	3	4	2	48	Verify driver current limit	3
Stepper Motor	Drives rotation of plate via friction mechanism	No function	Motor doesn't work; Rubber wheel doesn't work; Plate does not rotate; eggs will overlap and damage the yolk	Defective motor, broken winding, no motor power	4	4	4	96	Verify motor power	3
		Under function	Motor torque is too low; rubber wheel slips or moves slowly; Plate does not reach correct position	Low motor torque, poor friction contact, mechanical wear, overloaded plate	6	4	4	96	Improve friction contact and test motor torque under load	3
Rubber Wheel	Transfers motion to plate through friction contact	No function	Rubber wheel doesn't transfer motion; plate doesn't rotate; eggs may overlap and damage the yolk	Wheel missing, broken, loose, no contact with plate	2	4	1	8	Long-term service checks	3
Microswitch	Detects home position (0° reference)	No function	Plate position is not detected; system may drop egg at wrong plate position	Defective microswitch, broken wire	7	3	3	63	Verify switch position	2
		Intermittent function	Plate position signal is unreliable; plate may stop at wrong position	Defective microswitch, broken wire, poor mounting	7	4	4	112	Improve switch mounting	2
Base Plate	Structural base supporting entire mechanism	No function	Plate does not rotate; eggs fall on top of each other; yolk may be damaged	Plate jammed, blocked bearing, plate not aligned	6	3	2	36	Verify plate alignment	3

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